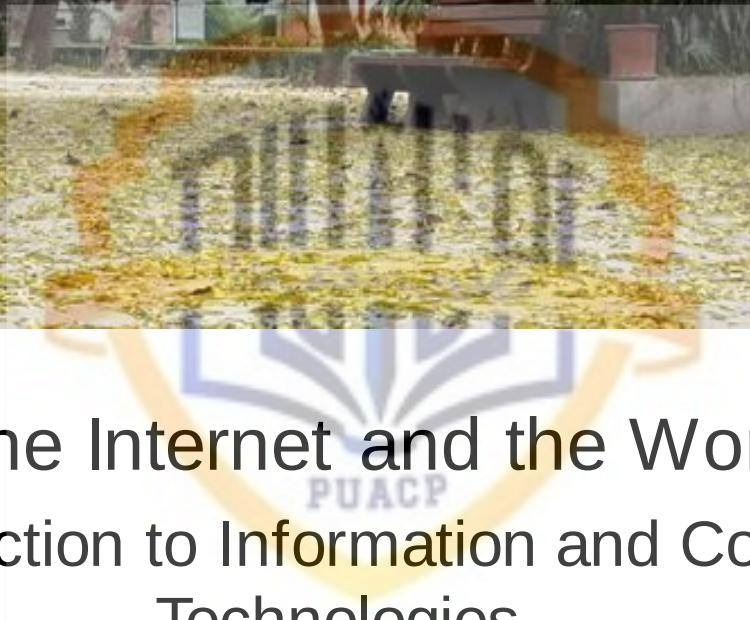




Punjab University College
of
Information Technology
“PUCIT”



Chapter 08 – The Internet and the World Wide Web

GE-161 Introduction to Information and Communication Technologies

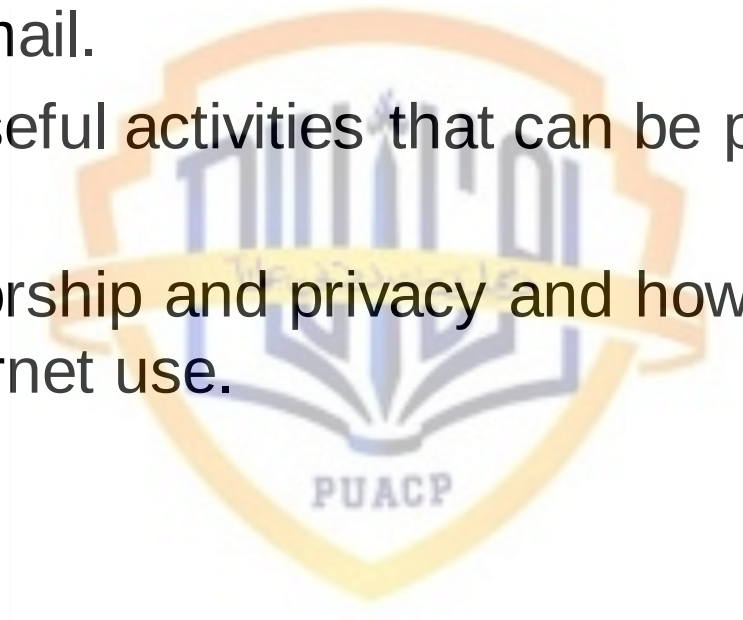
Prof. Dr. Syed Waqar ul Qounain
Department of Information Technology
University of the Punjab, Lahore

Learning Objectives

1. Discuss how the Internet evolved and what it is like today.
2. Identify the various types of individuals, companies, and organizations involved in the Internet community and explain their purposes.
3. Describe device and connection options for connecting to the Internet, as well as some considerations to keep in mind when selecting an ISP.
4. Understand how to search effectively for information on the Internet and how to cite Internet resources properly.

Learning Objectives

5. List several ways to communicate over the Internet, in addition to e-mail.
6. List several useful activities that can be performed via the Web.
7. Discuss censorship and privacy and how they are related to Internet use.



Overview

- This chapter covers:
 - The evolution of the Internet
 - The Internet community
 - Different options for connecting to the Internet
 - Internet searching
 - Common Internet applications available via the Internet
 - Societal issues that apply to Internet use

Evolution of the Internet

- Internet: Largest and most well-known computer network, linking millions of computers all over the world
 - Furnishes a standard way of obtaining information, disseminating information, accessing entertainment, and communicating with others
 - While *Internet* has become a household word only during the past two decades, it has actually operated in one form or another for much longer than that

Evolution of the Internet

- ARPANET: The predecessor of the Internet, named after the Advanced Research Projects Agency (ARPA), which sponsored its development
 - Initially connected four supercomputers
 - Eventually evolved into today's Internet
- The World Wide Web: The collection of Web pages available through the Internet
 - Proposed by Tim Berners-Lee in 1989
 - Originally only text-based content; release of the Mosaic browser in 1993 led to graphical content
 - Wide variety of content available via Web pages today (social networking, RSS, podcasts, blogs and wikis)
- Internet2: Researches advanced Internet applications and technologies

Evolution of the Internet

FIGURE 8-1
Using the Internet:
Back in the “old
days” versus now.



EARLY 1990s

Even at the beginning of the 1990s, using the Internet for most people meant learning how to work with a cryptic sequence of commands. Virtually all information was text-based.



TODAY

Today's Web organizes much of the Internet's content into easy-to-read pages that can contain text, graphics, animation, video, and interactive content that users access via hyperlinks.

The Internet Community Today

- Most members of the Internet community fall into one or more of the following groups:
 - Users: People who use the Internet
 - Internet service providers (ISPs): Provide access to the Internet, typically for a fee
 - Internet content providers: Provide Internet content
 - Businesses, non-profit organizations, educational institutions
 - Individuals

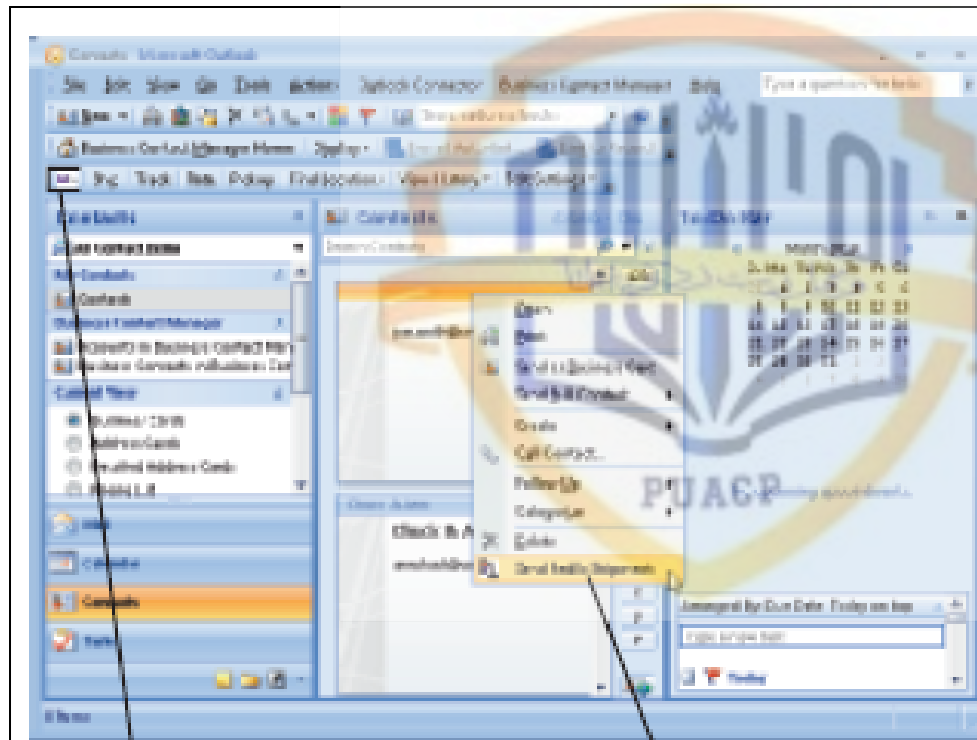
FIGURE 8-2
Companies that provide Internet access today include telephone, cable, and satellite companies.



The Internet Community Today

- Application service providers (ASPs): Companies that manage and distribute software-based services over the Internet
 - Web-based software, Software as a Service (SaaS), cloudware
 - Often fee-based business software
- Web service: Added to a Web page to provide specific services for end users
- Infrastructure companies: Own or operating the physical structure of the Internet
 - Conventional and mobile phone companies, cable companies, and satellite Internet providers

Web Services



The QuickShip toolbar can be used for a variety of shipping tasks.

Right-click a contact to create a shipment to that individual.

FIGURE 8-3
Web services. Once downloaded and installed, this Web service allows users to create and manage FedEx shipments from within Microsoft Outlook.

The Internet Community Today

- Hardware and software companies
 - Provide the hardware and software used in conjunction with the Internet and Web
- Government and other organizations
 - Some countries limit information and access
 - FCC influences communications
 - Internet Society (ISOC): Addresses issues impacting the future of the internet
 - Internet Corporation for Assigned Names and Numbers (ICANN): Domain and IP management
 - World Wide Web Consortium (W3C): Protocols and standards, ensures interoperability

Myths About the Internet

- Myth 1: The Internet is free
 - Most people and businesses pay for Internet access
 - Businesses, schools, and libraries lease communications lines from phone companies
 - Fee-based content is growing at a rapid pace
 - Music/movie downloads
 - Donation based sites



FIGURE 8-4
Fee-based Web content. The use of fee-based Web content, such as downloadable movies and music, is growing.

Myths About the Internet

- Myth 2: Someone controls the Internet
 - No single group or organization controls the Internet
 - Governments can regulate Internet use within its country, but difficult to enforce
- Myth 3: The Internet and World Wide Web are identical
 - Internet = physical network
 - WWW = one resource (Web pages) available via the Internet
 - Other resources are available via the Internet
 - e.g. FTP

Getting Set Up to Use the Internet

- Typically involves three decisions:
 - Determining the type of device you will use to access the Internet
 - Selecting the type of connection desired
 - Deciding on the Internet service provider to be used
- Once these determinations have been made, your computer can be set up to access the Internet

Type of Device

- Possible devices include:
 - Personal computers
 - Mobile phones
 - Gaming devices and televisions

FIGURE 8-5
A variety of devices
can be used to
access the Internet.



PERSONAL COMPUTERS



MOBILE PHONES



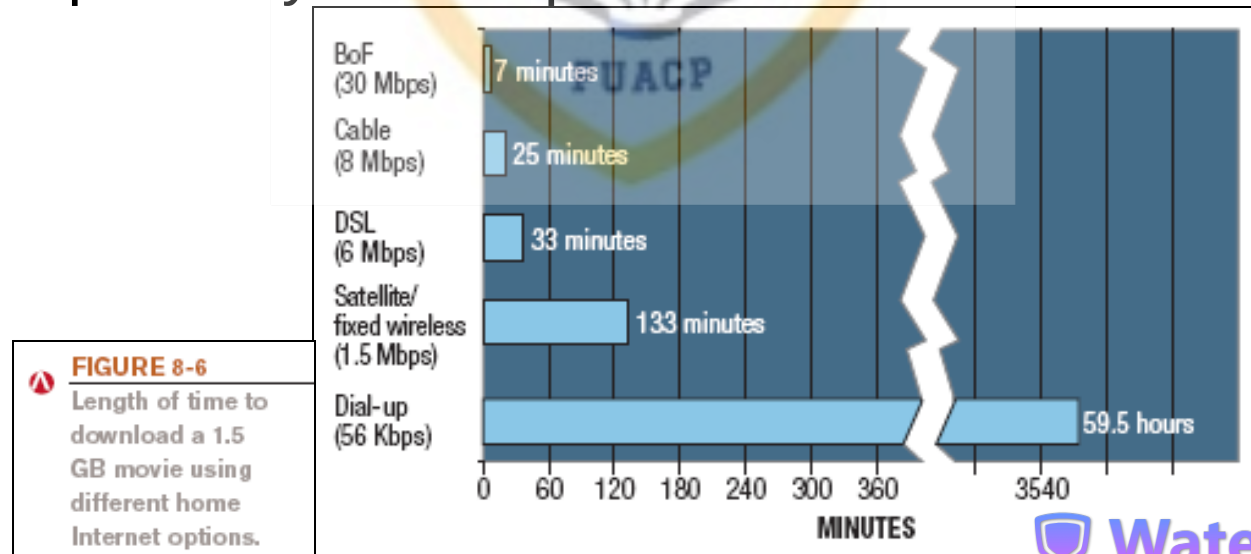
TELEVISIONS

Type of Connection and Internet Access

- Computer must be connected to access the Internet
 - Most connections today are broadband
- Connections can be:
 - Dial-up
 - Uses standard phone lines
 - Uses modem to dial-up ISP
 - Inconvenient
 - Slower, but cheaper
 - Ties up phone lines
 - Relatively secure from hackers

Type of Connection and Internet Access

- Direct (always on)
 - Device is continually connected to the Internet
 - Direct connections are typically broadband; fast speeds needed for many Web activities today
 - Because you are always connected, it is important to protect your computer from hackers



Direct Connections

- Cable: Most widely used home broadband connection
 - Fast, between 5 and 20 Mbps
 - Requires a cable modem
- DSL: Broadband delivered over telephone lines
 - Must be less than 3 miles from a switching station
 - Transmits over telephone lines but does not tie up the line
 - Typically 1-7 Mbps
- Satellite: Broadband option for rural areas
 - Slower and more expensive than cable or DSL
 - Available in many areas other broadband options are not
 - Requires satellite modem, and transceiver dish

Direct Connections

- Fixed wireless: Uses radio transmission towers rather than satellites
 - Not available in all areas
 - Uses Wi-Fi or WiMAX
- Broadband over Fiber (BoF): Delivers over fiber-optic cabling all the way to the building
 - Also called Fiber-to-the-premises (FTTP)
 - Fast, available in limited areas
 - Requires special networking equipment
- Mobile wireless: Access via mobile phone or device
- Hotspot: Public wireless networks (Wi-Fi)
 - Both free and fee-based are available

Type of Connection and Internet Access

TYPE OF INTERNET CONNECTION	AVAILABILITY	APPROXIMATE MAXIMUM SPEED*	APPROXIMATE MONTHLY PRICE
Conventional dial-up	Anywhere there is telephone service	56 Kbps	Free–\$20
Cable	Virtually anywhere cable TV service is available	5–20 Mbps	\$40–60
DSL	Within 3 miles of a switching station that supports DSL	1–7 Mbps	\$20–40
Satellite	Anywhere there is a clear view to the southern sky and where a satellite dish can be mounted and receive a signal	1–1.5 Mbps	\$60–80
Fixed wireless	Selected areas where service is available	1–2 Mbps	\$40–60
Broadband over fiber (BoF)	Anywhere fiber has been installed to the building	10–50 Mbps	\$50–145
Mobile wireless (3G)	Virtually anywhere cellular phone service is available	700 Kbps–1.7 Mbps	Varies greatly; often bundled with mobile phone service

* Download speed; most connections have slower upload speeds.

FIGURE 8-7
Typical home Internet connection options.

Wi-Fi Hotspots



FIGURE 8-9

Wi-Fi hotspots.

Hotspots are used to wirelessly connect to the Internet via the Internet connection belonging to a business, city, school, or other organization.



COFFEEHOUSES AND OTHER PUBLIC LOCATIONS

Often fee-based, though some are available for free.



HOSPITALS, BUSINESSES, AND OTHER ORGANIZATIONS

Usually designed for employees but are sometimes also available free to visitors.



COLLEGE CAMPUSES

Usually designed for students and faculty; sometimes used directly in class for student assignments, as shown here.

Selecting an ISP

AREA	QUESTIONS TO ASK
Services	Is the service compatible with my device? Is there a monthly bandwidth limit? How many e-mail addresses can I have? What is the size limit on incoming and outgoing e-mail messages and attachments? Do I have a choice between conventional and Web-based e-mail? Is there dial-up service that I can use when I'm away from home? Are there any special member features or benefits? Does the service include Web site hosting?
Speed	How fast are the maximum and usual downstream (ISP to my PC) speeds? How fast are the maximum and usual upstream (my PC to ISP) speeds? How much does the service slow down under adverse conditions, such as high traffic or poor weather?
Support	Is 24/7 telephone technical support available? Is Web-based technical support (such as via e-mail) available? Is there ever a charge for technical support?
Cost	What is the monthly cost for the service? Is it lower if I prepay a few months in advance? Are different tiers available? Is there a set-up fee? If so, can it be waived with a 6-month or 12-month agreement? What is the cost of any additional hardware needed, such as modem or transceiver? Can the fee be waived with a long-term service agreement? Are there any other services (telephone service, or cable or satellite TV, for instance) available from this provider that I have or want and that can be combined with Internet access for a lower total cost?



FIGURE 8-10

Choosing an ISP.

Some questions to ask before making your final selection.



Setting Up Your Computer

- To set up your computer
 - Install necessary hardware
 - Install necessary software
 - Select user name, access telephone number (for dial-up connections), and a payment method
 - Perform any necessary set-up to share the connection with other computers or devices

Quick Quiz

1. The Internet began as an experimental network known by which name?
 - a. ARPANET
 - b. Internet2
 - c. World Wide Web
2. True or False: Cable is a common type of direct Internet connection.
3. A type of always-on broadband Internet access available through conventional phone lines that does not tie up your phone line is _____.

Answers:

1) a; 2) True; 3) DSL

Searching the Internet

- Effective Internet searching: A very important skill
- Search sites: Web sites specifically designed to help users search for Web pages that match specified keywords or selected categories
 - Typically use a search engine in conjunction with a database containing information about Web pages to locate appropriate Web pages
 - Many search sites available (Google, Bing, Yahoo, Ask.com, Cuil, etc)
 - Real-time search engines are emerging
 - Search the Web live rather than relying on a database

Searching the Internet

- Keyword search
 - Keywords are typed in a search box to locate information on the Internet
 - Matching Web pages are called hits
 - Clicking on a Web page name displays that page
- Directory search
 - Categories are selected to locate information on the Internet
- Many search sites contain additional tools
 - Search for music files, image files, newsgroups, news articles, maps, people, telephone numbers
 - Google is one of the most versatile search sites

Searching the Internet



FIGURE 8-11

Using a search site.

Search Site Tools

FUNCTION	EXPLANATION
Calculator	Enter a mathematical expression or a conversion to see the result.
Currency converter	Enter an amount and currency type to see the corresponding value.
Dictionary	Enter the term <i>define</i> followed by a term to view definitions for that term from online sources.
Flight information	Enter an airline and a flight number to see status information.
Movie showtimes	Enter the term <i>movie</i> followed by a ZIP code to view movies showing in that area.
Number search	Enter a UPS, FedEx, or USPS tracking number; an area code; or a UPC code to view the associated information.
Phonebook	Enter a name followed by a city and a state, a ZIP code, or an area code to look up that person's address and phone number.
Reverse phonebook	Enter a telephone number to look up the person or business associated with that number.
Stock quotes	Enter one or more stock ticker symbols to retrieve stock quotes.
Street maps	Enter an address to find a map to that location.
Time	Enter the term <i>time</i> followed by a city name to see the current time in that city.
Travel conditions	Enter an airport code followed by the term <i>airport</i> to view current conditions at that airport.
Weather	Enter the term <i>weather</i> followed by a city name or ZIP code to view the weather for that location.
Yellow pages	Enter a type of business and city name or ZIP code to view businesses in that local area.

EXAMPLES:

10 miles = 52 800 feet

**10:10pm Wednesday (CEST)
Time in Paris, France**

Weather for San Francisco, CA
68°F
 Mostly Cloudy
 Wind: NW at 1 mph
 Humidity: 76%

Wed 66°F 50°F	Thu 68°F 52°F	Fri 70°F 52°F
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FIGURE 8-13
Google search tools

Search Strategies

- Search strategies:
 - Phrase searching (more than one keyword)

SEARCH PHRASE USED	SEARCH SITE	NUMBER OF PAGES FOUND	TITLE OF FIRST TWO NONSPONSORED PAGES FOUND*
dogs	Google	48,600,000	Dogs & Puppies – Next Day Pets Dogs – Dog Information, Pictures and Reviews for over 350 Dogs
	Yahoo!	633,000,000	American Kennel Club (AKC) Animal Planet's Dog Guide
hand signals	Google	58,600,000	Hand Signals Bicycle Safety – Hand Signals
	Yahoo!	45,100,000	Hand Signals – Wikipedia, the free encyclopedia Bicycle Safety – Hand Signals
dog hand signals	Google	1,010,000	DDEAF Training Hand Signals How to Teach a Dog Hand Signals eHow.com
	Yahoo!	7,660,000	How to Train a Dog Using Hand Signals eHow.com Dog Training – Hand Signals – Amazingdogtrainingman.com
dog hand signals	Google	713	How to Teach a Dog Hand Signals eHow.com D for Dog – Deaf dog hand signals sign language
	Yahoo!	213	How to Train a Dog Using Hand Signals eHow.com Dog Training – Hand Signals – Amazingdogtrainingman.com

* Highlighted entries indicate Web pages about dog hand signals.

FIGURE 8-14
Examples of phrase searching. Using different search phrases and different search sites can dramatically change the search results.

Search Strategies

- Boolean operators (AND, OR, NOT)
- Multiple search sites
- Appropriate keywords
- Synonyms (different words that mean the same thing)
- Variant word forms (variations of your keywords, alternate spellings, etc.)
- Wildcards (such as * to search for keyword patterns)
- Field searches (searches limited to a particular characteristic, such as page title, page text, URL, top level domain or Web site)

Search Strategies

FIELD TYPE	EXAMPLE	EXPLANATION
Title	title:"tax tips"	Searches for Web pages containing the words "tax tips" in the page title.
URL	url:taxtips	Searches for Web pages containing "taxtips" in the page URL.
Text	text:"tax tips"	Searches for Web pages containing "tax tips" in the text of the page.
Site	forms site:irs.gov	Searches for Web pages associated with the keyword "forms" that are located only on the irs.gov Web site.
Domain	tax tips site:*.gov	Searches for Web pages associated with the keywords "tax tips" that are located on government Web sites (they can have anything for the first part of the domain name, but must have a .gov TLD).

FIGURE 8-15
Field searching.
Field searches limit search results to just those pages that match specific field criteria, in addition to any specified search criteria.

Evaluating Search Results

- Evaluating search results:
 - Does the title and listed description sound appropriate for the information you are seeking?
 - Is the URL from an appropriate company or organization?
 - You should also evaluate:
 - The author
 - The source
 - Determine if reliable or biased
 - The date
 - Many online articles are years old
 - Verify online information with a second source

Citing Internet Resources

- To avoid plagiarism, proper citation procedures should be used for all Internet content used in a paper, book, or on a Web site
- Plagiarism is discussed in more detail in Chapter 16

TYPE OF RESOURCE	CITATION EXAMPLE
Web page article (magazine)	Dvorak, John (2009, May 5). Data Mining and the Death of Privacy. <i>PC Magazine</i> . Retrieved February 14, 2010, from http://www.pcmag.com/article2/0,1895,2346287,00.asp
Web page article (journal)	Sensmeier, Joyce (2009, March). Advancing the EHR (Electronic Health Record): Are We There Yet? <i>Nursing Management</i> , 40 no. 3. Retrieved June 21, 2010, from http://www.nursingcenter.com/library/JournalArticle.asp?Article_ID=850211
Web page article (not appearing in a periodical)	Baldor, Lolita (2009, May 6). Air Traffic Systems Vulnerable to Cyber Attack. <i>MSNBC</i> . Retrieved March 15, 2010, from http://www.msnbc.msn.com/id/30602242
Web page content (not an article)	<i>Browse the Web Safely</i> . (n.d.) Retrieved April 11, 2010 from http://www.symantec.com/norton/security_response/browsewebsafely.jsp
E-mail (cited in text, not reference list)	Maria Rodriguez (personal communication, March 28, 2010).

 **FIGURE 8-16**
Citing Web sources.

Quick Quiz

1. With which of the following does the user select a category matching the topic for which he or she is searching?
 - a. keyword search
 - b. directory search
 - c. field search
2. True or False: The search phrase “trailer NOT horse” would locate Web pages about horse trailers
3. The program used by many search sites to retrieve matching Web pages from their database is called a(n) _____.

Answers:

1) b; 2) False; 3) search engine

Beyond Browsing and E-Mail

- Many other activities available via the Internet in addition to Web browsing and e-mail
- Types of online communications
 - Instant messaging (IM): Exchanging real-time messages
 - Typically real-time (presence technology)
 - Often can be used for a variety of other tasks (voice calls, multiplayer games, etc.)
 - Unified communications (UC)
 - Text messaging

Types of Online Communications

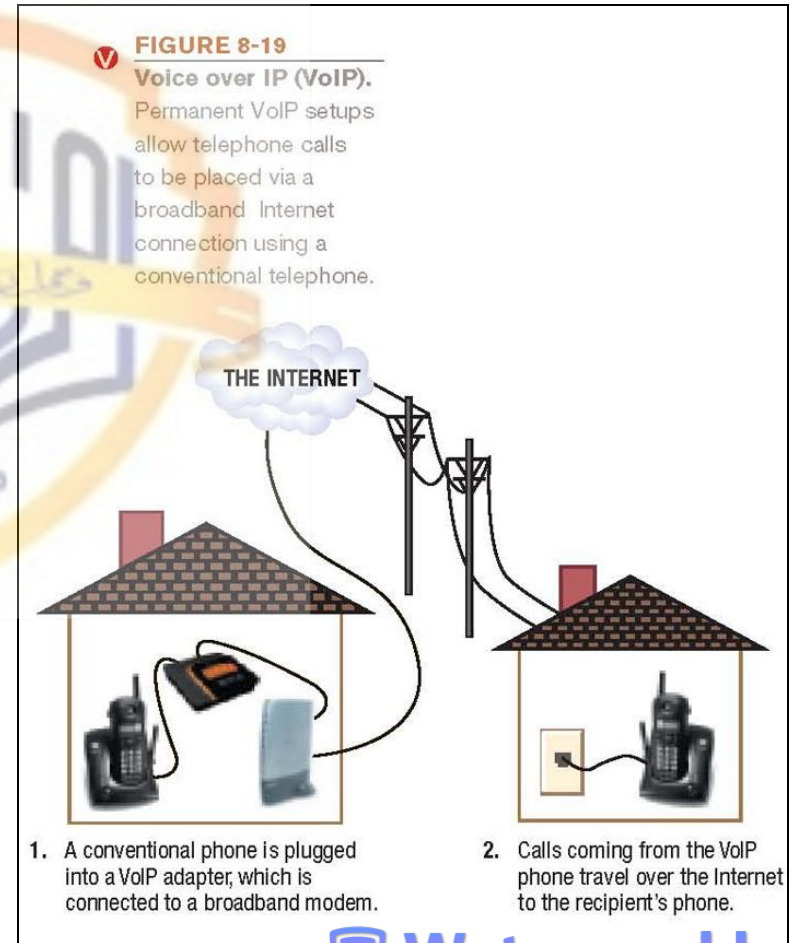
- Twittering (microblogging)
 - Send and receive tweets
 - Used in both business and personal lives
 - Other types of status updates are available via some social networking sites



FIGURE 8-18
Twitter. Allows individuals to post tweets via computer, mobile phone, or Xbox (as shown here).

Types of Online Communications

- Message boards: Post and read messages
- Voice over Internet Protocol (VoIP): Making telephone calls over the Internet
 - Computer to computer
 - Skype, IM, etc.
 - More permanent replaces landline phone
 - Does not work when Internet connection or power is out



Types of Online Communications

- Web conference: Face-to-face meeting taking place via the Web
 - Typically uses video cameras and microphones to enable participants to see and hear each other
 - Via IM (personal use) or Web conference service (business use)
- Webinar: A seminar presented via the Web.



FIGURE 8-20
Web conferencing.
Allows individuals to talk with and see each other in real time, such as this student talking to her family from her dorm room.

Social Networking

- Social networking site: A site that enables a community of individuals to communicate and share information
 - MySpace, Facebook, etc. allow people to post information about themselves
 - Video and photo sharing (YouTube, Flickr, etc.)
 - Used by schools for classmates to meet and share
 - Used in politics and business
 - Used by families
- Users should be careful not to reveal too much about themselves for safety reasons

Social Networking



PERSONAL PROFILING SITES
Allow individuals to post information about themselves, link pages with friends, exchange messages, and so forth.



BUSINESS NETWORKING SITES
Help businesspeople find business contacts, potential new employees and clients, dinner and traveling partners during business trips, and so forth.

FIGURE 8-21
Social networking sites. A variety of social networking sites are available to meet different needs.

Online Shopping and Investing

- E-commerce: Online financial transactions
- Precautions
 - Be sure all e-commerce activities are performed via a secure Web server (<https://>)
 - Use a credit card or online payment service whenever possible
- Online shopping: Buying products or services online
 - Can purchase items via manufacturers, large businesses, small businesses, etc.
 - Usually paid via credit card, though other options may be available

Online Shopping and Investing

- Online auctions: Bids are placed for items, and the highest bidder purchases the item
 - Sellers list items (such as via eBay)
 - Can bid until auction closes
 - Winning bidder arranges payments and delivery with seller
 - Seller pays auction fees
- Online banking: Performing banking activities via the Web (transfer funds, check balances, pay bills, etc.)
- Online investing: Buying and selling stocks or other types of investments via the Web
 - Online portfolios

Online Shopping and Investing



FIGURE 8-22

Online shopping.

Allows you to purchase goods and services online.

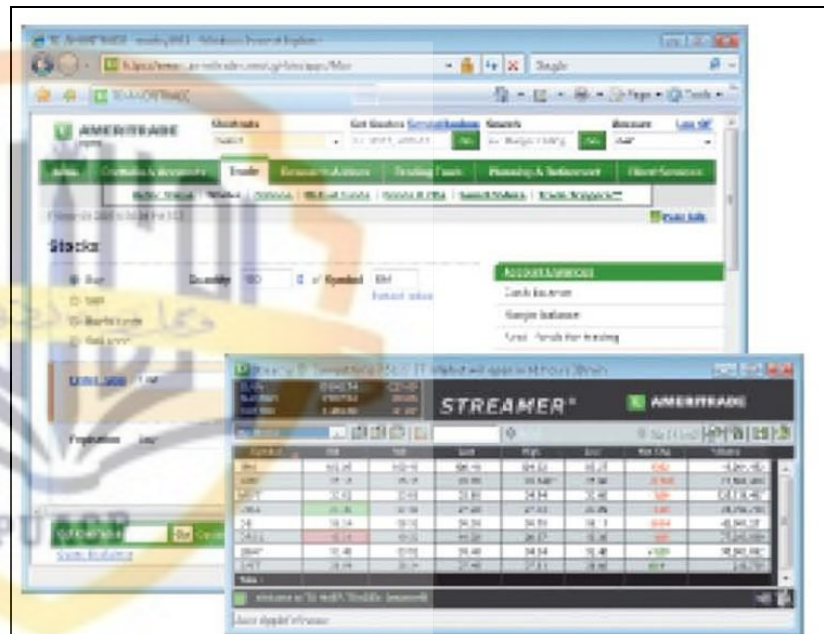


FIGURE 8-23

Online investing.

Allows you to buy and sell stock, view your portfolio, get real-time quotes, and more.

Online Entertainment

- Online music: Music played or obtained via the Web
 - Internet radio stations: Listen to music live
 - Online music stores: Used to purchase and download music singles and albums in digital format legally
- Online video: Video watched or downloaded via the Web
- Online TV: Live or recorded TV shows available via the Web
- Online movies: Feature films available via the Web
- Video-on-demand (VoD): The process of downloading movies and television shows, on demand, via the Web
- Online gaming: Playing games via the Web
 - Web-based games, online multiplayer games, etc.

Online Entertainment

FIGURE 8-24
Online TV and movies.



MOBILE TV

Both live and recorded TV shows can be delivered directly to a mobile phone.



ONLINE TV AND MOVIES

TV shows and movies can be watched online for free via a variety of Web sites.



VIDEO-ON-DEMAND (VOD)

Fee-based feature films and other content can be delivered to a computer or TV.

Online News, Reference, and Information

- News sites
 - Available through Web sites belonging to news organizations, television networks, newspapers, magazines, etc.
 - Usually updated throughout the day
 - News archives are often available (sometimes requires a fee)
- Reference sites
 - Provide access to specific types of useful information
 - Phone and address directories, weather, maps, home values, encyclopedias, dictionaries, etc.

Online News, Reference, and Information



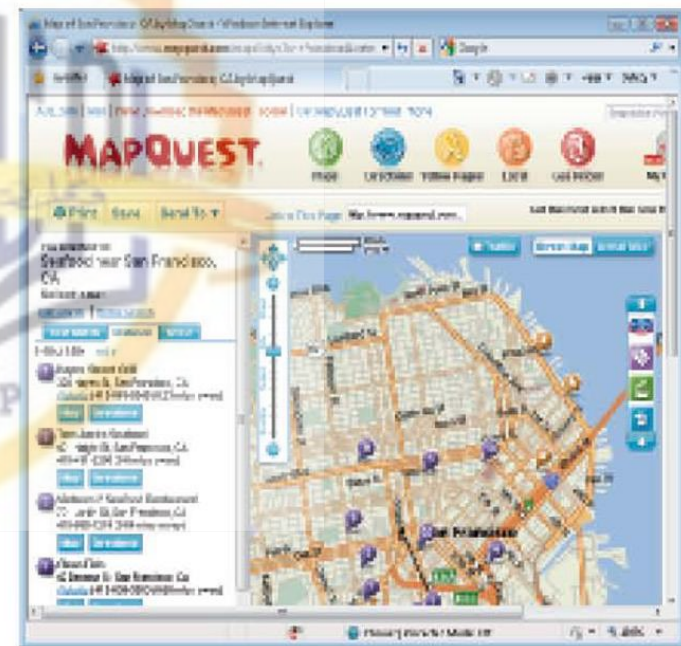
FIGURE 8-25

Online news and reference.



NEWS SITES

News organizations typically update their sites several times per day to provide access to the most current news and information.



REFERENCE SITES

Reference Web sites provide access to specific types of useful information, such as the maps and driving directions available via this Web site.

Online News, Reference, and Information

- Portal Web page: Designed to be displayed as a browser's home page, typically can be customized
 - Often contain news, calendars, e-mail, and other tools
 - Usually personalized
 - iGoogle, My Yahoo!, My MSN, etc.



Online News, Reference, and Information

- RSS (Really Simple Syndication): Delivers selected Web content to subscribers as the content is published to a Web site
 - Individuals subscribe to an RSS feed and new content is delivered to them as it becomes available
- Podcasts: Recorded audio or video file that can be played or downloaded via the Web
 - Prepared by individuals and businesses
 - Used to share knowledge, express opinions, share original poems, songs, or short stories
 - Typically uploaded to the Web on a regular basis

Product, Corporate, and Government Information

- Vast amount of product and government information available online
- Sites dedicated to provide information about politics, social causes, etc.



FIGURE 8-27
FactCheck.org. This Web site can be used to check the accuracy of political statements.

Online Education and Writing

- Online education: Using the Internet to facilitate learning
 - Web-based training (WBT): Instruction delivered on an individual basis via the Web
 - Corporate training, tutorials, distance learning, etc.



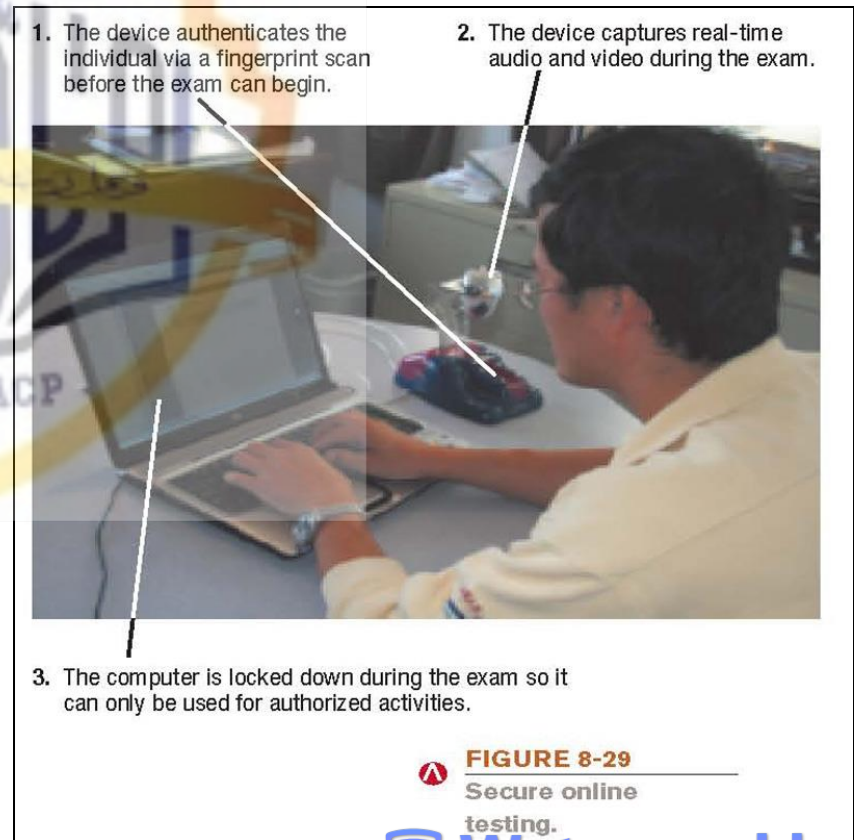
FIGURE 8-28
Web-based learning.
The Understanding Computers Web site provides a variety of Web-based learning opportunities for readers of this textbook.

Online Education and Writing

- Distance learning: When the student is physically located away from the instructor and other students; instruction take place via the Web
 - Advantages: Self-paced instruction, flexible location, up-to-date material, immediate feedback, and customized content
 - Disadvantages: Technology requirements and problems, anonymity, and lack of face-to-face contact

Online Education and Writing

- Online testing: Taking tests via the Internet
 - Available for both objective and performance-based exams
 - Typically are graded automatically
 - Security is an issue
 - Authorized testing centers
 - Secure testing systems



Online Education and Writing

– Online writing

- Blogs: A Web page that contains short, frequently updated entries in chronological order, typically by just one individual
- Wikis: A collaborative Web page that is designed to be edited and republished by a variety of individuals
 - Wikipedia is largest
 - » Carefully evaluate content, as irresponsible individuals can enter erroneous information
- E-portfolios: A collection of an individual's work accessible via the Web

Online Education and Writing

FIGURE 8-30

Blogs. Allow individuals to post entries to an online personal journal.



FIGURE 8-31

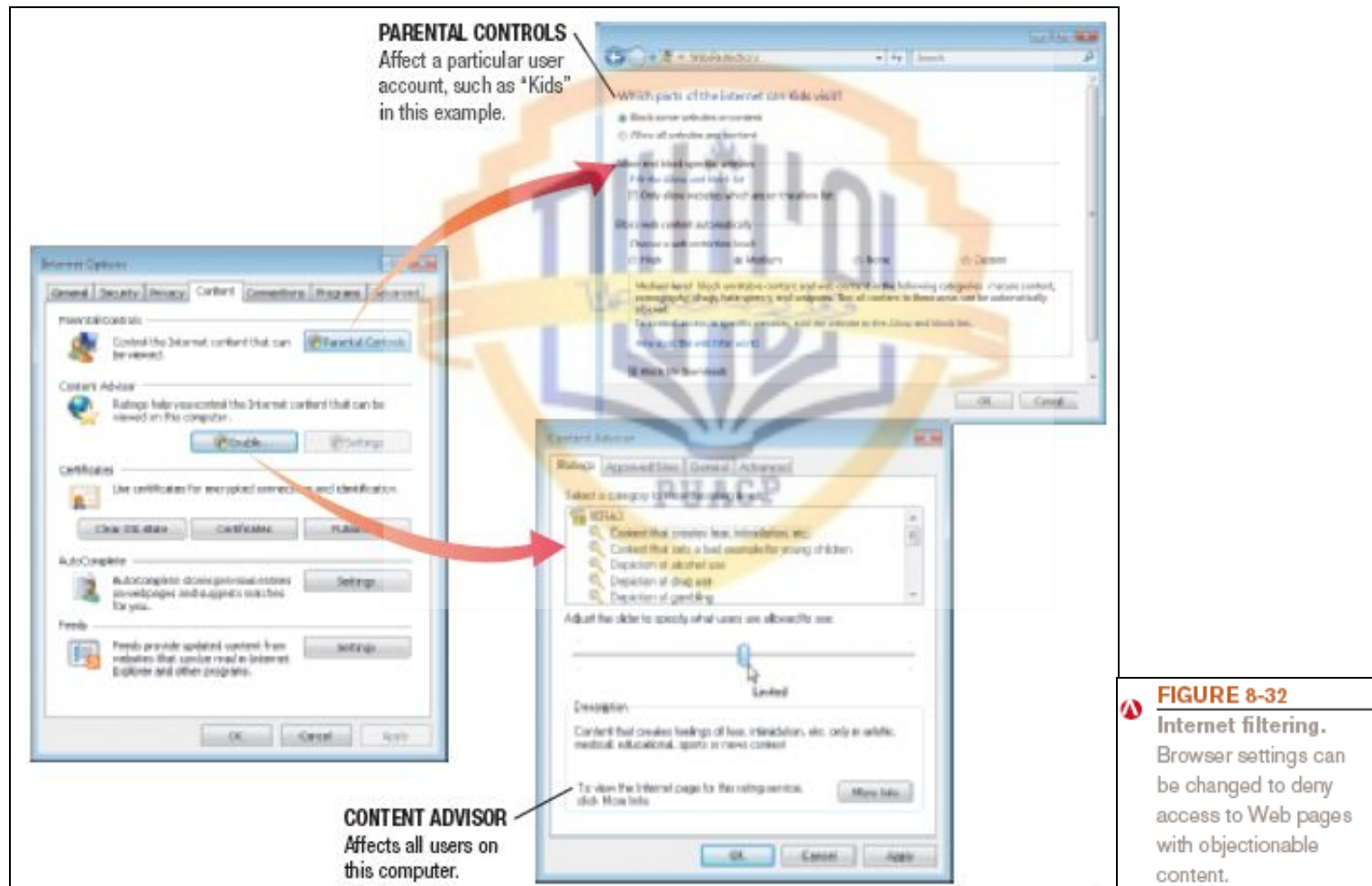
Wikis. Wikis, such as the Wikipedia collaborative online encyclopedia shown here, can be edited by any authorized individual.



Censorship and Privacy Issues

- Censorship issues
 - Some countries block some Internet content
 - Free speech vs. offensive or indecent materials
 - Internet filtering: Using software or browser options to block access to particular Web pages or types of Web pages
 - Used by individuals, schools, employers, public computers, etc.
 - Can use browser settings or special filtering software

Internet Filtering



Censorship and Privacy Issues

- Web browsing privacy: Privacy of Web activities
 - Cookies: Small files stored on the user's hard drive by a Web server
 - Add functionality (such as shopping carts and personal preferences) to Web sites
 - Can be used to track Web activity
 - Cookie information can be personally identifiable or non-personally identifiable
 - Cookie data can be viewed or deleted
 - Cookie settings can be changed and software can be used to manage cookies

Cookies

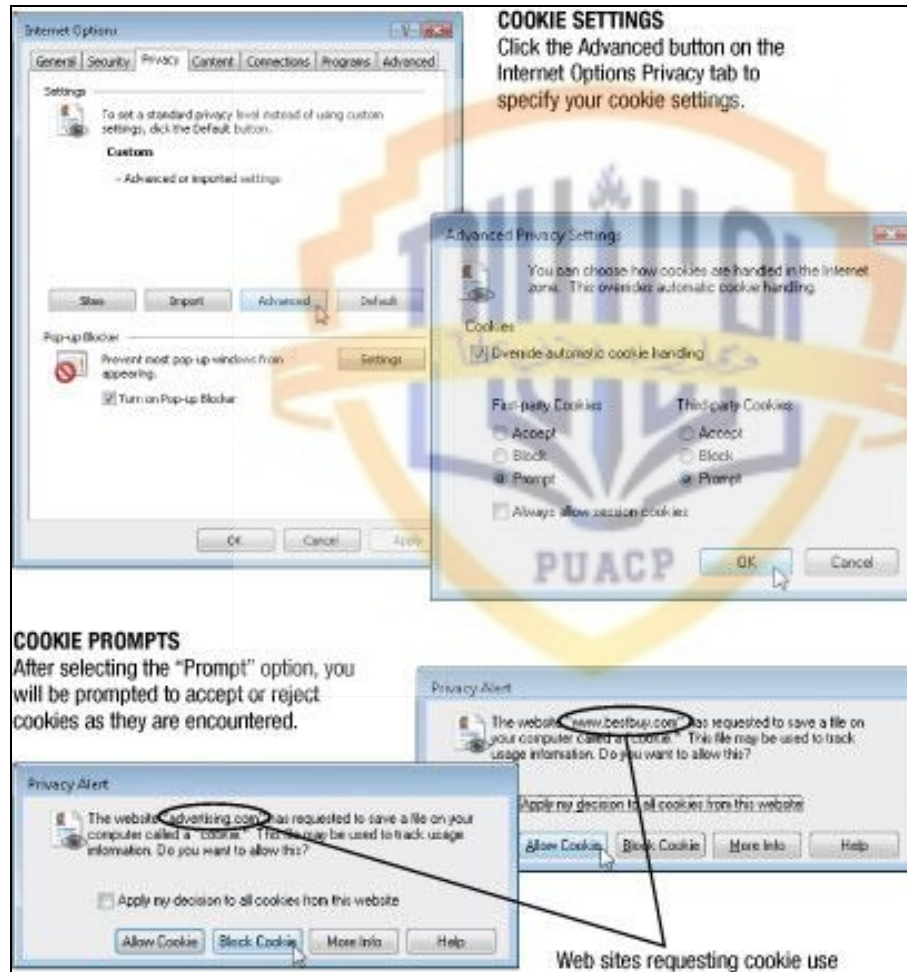


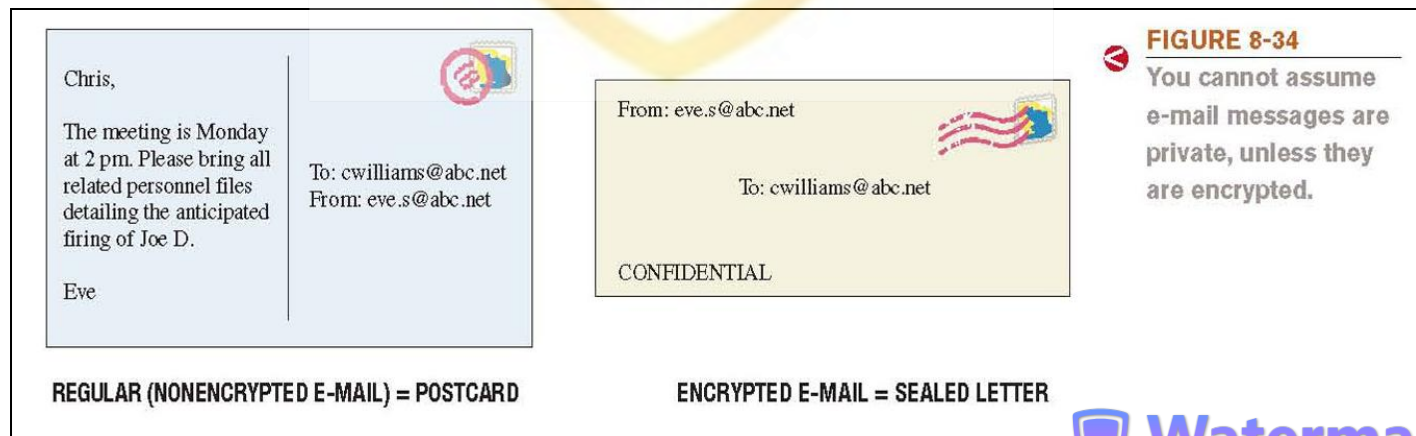
FIGURE 8-33
Browser cookie management. The desired cookies settings can be specified in the browser settings.

Censorship and Privacy Issues

- Spyware: Software installed without users knowledge that transmits data secretly through the user's Internet connection
 - Sometimes used by advertisers to gather marketing information
 - Used by criminals to gather personal data stored on your computer
 - Can be blocked and/or removed using security software
- Adware: Software supported by onscreen advertising
 - Often included in free programs
 - Does not gather information

Censorship and Privacy Issues

- E-mail privacy: Privacy of the e-mail messages you send and receive
 - Only encrypted e-mail can be transmitted privately
 - Employers and ISPs have access to the e-mail you send through those organizations
 - Businesses and ISPs typically archive e-mail messages



Quick Quiz

1. Ordering a movie or television show to be downloaded from the Internet is referred to as?
 - a. Interactive TV
 - b. Online radio
 - c. Video-on-demand (VOD)
2. True or False: A URL that begins with *https://* indicates the page being viewed is secure.
3. Blocking access to particular Web pages or types of Web pages using browser settings or special software is called Internet

Answers:

1) c; 2) True; 3) filtering

Summary

- Evolution of the Internet
- Getting Set Up to Use the Internet
- Searching the Internet
- Beyond Browsing and E-Mail
- Censorship and Privacy Issues

